

Band-Limitedness of the Hardy Z-Function in the Phase Variable

Stephen Crowley

April 19, 2026

Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the Hardy Z-function, and $\theta(t) = \text{Im} \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta function. The function Z is meromorphic on $\mathbb{C}$ with poles inherited from ζ ; on the open strip $\Omega := \{|\text{Im } t| < \frac{1}{2}\}$, Z is holomorphic. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi)$; for any constant $c > c_*$, define $\Theta(t) := \theta(t) + ct$. Then $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing bijection, and the unitary inverse phase pullback

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (1)$$

is a real-valued function on $\mathbb{R}$ whose Fourier support is contained in $[-1, 1]$. By the Paley–Wiener theorem, Y extends to an entire function of exponential type ≤ 1 . Combined with the non-negativity of the spectral measure $S := |\hat{Y}|^2$ and Akhiezer’s theorem, Y belongs to the Laguerre–Pólya class, so all zeros of Y are real. The sub-strip identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$, established on each sub-strip $\Omega_\varepsilon := \{|\text{Im } t| < \frac{1}{2} - \varepsilon\}$ for $\varepsilon \in (0, \frac{1}{2})$ under an ε -dependent lower bound on c , transfers the real zero locus of Y to that of Z on Ω_ε . Taking the union over ε gives reality of zeros of Z on Ω . Therefore every non-trivial zero ρ of ζ satisfies $\text{Re } \rho = \frac{1}{2}$.

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-valued on $\mathbb{R}$ and holomorphic on the open strip $\{|\text{Im } t| < \frac{1}{2}\}$. It is not entire: the pole of ζ at $s = 1$ pulls back under $s = \frac{1}{2} + it$ to a pole at $t = -i/2$. The zero locus of Z in the open strip $\{|\text{Im } t| < \frac{1}{2}\}$ determines the zero locus of ζ in the critical strip $\{0 < \text{Re } s < 1\}$ via $s = \frac{1}{2} + it$.

The Riemann–Siegel phase θ fails to be monotone on $[0, \infty)$: θ' is strictly increasing but negative on $[0, t_c]$ and positive on $[t_c, \infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ' . The argument presented here restores monotonicity on $\mathbb{R}$ by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, ensuring $\Theta' > 0$ on $\mathbb{R}$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ bijective. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$, defined via this real bijection, extends to an entire function whose zeros are real. The sub-strip identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is established on each $\Omega_\varepsilon := \{|\text{Im } t| < \frac{1}{2} - \varepsilon\}$ ($\varepsilon \in (0, \frac{1}{2})$) under an ε -dependent lower bound on c , and the union over ε covers Ω .

All spectral quantities, in particular the mode thresholds β_n , are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y : $\text{supp } S \subseteq [-1, 1]$. Paley–Wiener gives Y entire of exponential type ≤ 1 ; Akhiezer gives reality of its zeros; the unitary identity transfers reality to the zeros of Z inside the open strip; the equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, valid on $\{|\text{Im } t| < \frac{1}{2}\} \setminus \{\pm i/2\}$ where the non-trivial zeros of ζ lie, yields the conclusion for ζ .

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad (2)$$

$$\theta'(t) = \frac{1}{2} \operatorname{Re} \psi\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{1}{2} \log \pi, \quad (3)$$

$$\theta''(t) = -\frac{1}{4} \operatorname{Im} \psi'\left(\frac{1}{4} + \frac{it}{2}\right) \quad (4)$$

The poles of ψ and ψ' at $\frac{1}{4} + \frac{it}{2} \in \mathbb{Z}_{\leq 0}$ correspond to $t \in \{-i(\frac{1}{2} + 2k) : k \geq 0\}$; the nearest to $\mathbb{R}$ is $t = -i/2$, at distance $1/2$. Therefore $\theta, \theta', \theta''$ are holomorphic on the open strip $\{|\operatorname{Im} t| < \frac{1}{2}\}$. For any $\varepsilon \in (0, \frac{1}{2})$, θ' and θ'' are bounded on the closed sub-strip $\{|\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon\}$.

Lemma 1 (Positivity of θ''). *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k \geq 0} (s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \quad (k \geq 0, t > 0) \quad (5)$$

Summing, $\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. $\square$

Lemma 2 (Exact value of $\theta'(0)$).

$$\theta'(0) = \frac{1}{2} \psi\left(\frac{1}{4}\right) - \frac{1}{2} \log \pi = -\frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi) \quad (6)$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3 \log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6 \log 2$ ([1], §6.3.4). $\square$

Definition 1 (Admissible constant at level ε). *For $\varepsilon \in (0, \frac{1}{2})$,*

$$c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3 \log 2 + \frac{\pi}{2} + \log \pi), \quad (7)$$

$$M_\varepsilon := \sup_{|\operatorname{Im} t| \leq 1/2 - \varepsilon} |\theta''(t)| = \frac{1}{4} \sup_{|\operatorname{Im} t| \leq 1/2 - \varepsilon} |\operatorname{Im} \psi'(\frac{1}{4} + \frac{it}{2})|, \quad (8)$$

$$c_{**}(\varepsilon) := c_* + M_\varepsilon. \quad (9)$$

A real c is ε -admissible if $c > c_{**}(\varepsilon)$.

Lemma 3 (Finiteness of M_ε). *$M_\varepsilon < \infty$ for every $\varepsilon \in (0, \frac{1}{2})$.*

Proof. ψ' is holomorphic on $\mathbb{C} \setminus \mathbb{Z}_{\leq 0}$. For $\varepsilon \in (0, \frac{1}{2})$, the closed set $\{\frac{1}{4} + \frac{it}{2} : |\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon\}$ has real part in $[\frac{\varepsilon}{2}, \frac{1}{2} - \frac{\varepsilon}{2}]$, and so is at distance $\geq \frac{\varepsilon}{2}$ from $\mathbb{Z}_{\leq 0}$. On the compact rectangle $\{|\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon, |\operatorname{Re} t| \leq R\}$, $|\theta''|$ is continuous and bounded; outside, $|\theta''| = O(|\operatorname{Re} t|^{-1}) \rightarrow 0$ by the trigamma asymptotic $\psi'(s) = O(|s|^{-1})$. The sup is therefore finite. $\square$

Lemma 4 (Strict positivity of Θ' on the ε -sub-strip). *For every $\varepsilon \in (0, \frac{1}{2})$ and every ε -admissible c ,*

$$\operatorname{Re} \Theta'(t) \geq c - c_{**}(\varepsilon) > 0 \quad \text{on } \{|\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon\}.$$

Proof. For $t = \tau + i\sigma$ with $|\sigma| \leq \frac{1}{2} - \varepsilon$,

$$\theta'(\tau + i\sigma) - \theta'(\tau) = \int_0^\sigma i\theta''(\tau + is) ds,$$

so $|\operatorname{Re} \theta'(\tau + i\sigma) - \theta'(\tau)| \leq \int_0^{|\sigma|} |\theta''(\tau + is)| ds \leq |\sigma| M_\varepsilon \leq (\frac{1}{2} - \varepsilon) M_\varepsilon \leq M_\varepsilon$. θ' is even on $\mathbb{R}$ and increasing on $[0, \infty)$ (Lemma 1), so $\theta'(\tau) \geq \theta'(0) = -c_*$. Therefore $\operatorname{Re} \Theta'(t) = \operatorname{Re} \theta'(t) + c \geq -c_* - M_\varepsilon + c = c - c_{**}(\varepsilon) > 0$. $\square$

Lemma 5 (Injectivity of Θ on the ε -sub-strip). *For every $\varepsilon \in (0, \frac{1}{2})$ and every ε -admissible c , Θ is injective on $\{|\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon\}$. Moreover, for every admissible c in the sense that $c > c_*$, Θ is bijective $\mathbb{R} \rightarrow \mathbb{R}$.*

Proof. If $\Theta(t_1) = \Theta(t_2)$ with $t_1, t_2 \in \{|\operatorname{Im} t| \leq \frac{1}{2} - \varepsilon\}$, the straight line $\gamma(s) = t_1 + s(t_2 - t_1)$ lies in the same convex closed sub-strip, and $\Theta(t_2) - \Theta(t_1) = (t_2 - t_1) \int_0^1 \Theta'(\gamma(s)) ds$. The real part of the integral is $\geq c - c_{**}(\varepsilon) > 0$ by Lemma 4, so the integral is nonzero, forcing $t_1 = t_2$. On $\mathbb{R}$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ for any $c > c_*$, so Θ is strictly increasing on $\mathbb{R}$; combined with $\Theta(\tau) \rightarrow \pm\infty$ as $\tau \rightarrow \pm\infty$ (from $\theta(\tau) \sim \frac{\tau}{2} \log(\tau/2\pi)$), $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is bijective. $\square$

3 The Hardy Z-Function on the Strip

Lemma 6 (Holomorphy of Z on the strip). *$\zeta(\frac{1}{2} + it)$ is meromorphic on $\mathbb{C}$ with a single simple pole at $t = -i/2$ (image of $s = 1$ under $s = \frac{1}{2} + it$). The factor $e^{i\theta(t)}$ is holomorphic and non-vanishing on $\{|\operatorname{Im} t| < 1\}$. Therefore Z is meromorphic on $\{|\operatorname{Im} t| < 1\}$ and holomorphic on*

$$\Omega := \{|\operatorname{Im} t| < \tfrac{1}{2}\} \quad (10)$$

Proof. Meromorphy of ζ with simple pole at $s = 1$ is classical ([8], §2.1). The pullback to the t -plane places the pole at $t = -i/2$, on the boundary of Ω . The factor $e^{i\theta(t)}$ is holomorphic and non-vanishing on Ω since θ is holomorphic on Ω by (2). $\square$

4 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For $c > c_*$ and $u \geq 0$, the unitary inverse phase pullback of Z is*

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \quad (11)$$

Extend to $\mathbb{R}$ by $Y(-u) := Y(u)$. The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ from Lemma 5, and $\Theta'(\Theta^{-1}(u)) > 0$ on $\mathbb{R}$ since $\Theta'(\tau) \geq c - c_ > 0$ on $\mathbb{R}$. The half-Jacobian factor renders the map $Z \mapsto Y$ an L^2 -isometry under the change of variable $u = \Theta(t)$.*

Y is real-valued on $\mathbb{R}$.

5 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n := \Theta'(2\pi n^2) = \tfrac{1}{2} \operatorname{Re} \psi\left(\tfrac{1}{4} + i\pi n^2\right) - \tfrac{1}{2} \log \pi + c \quad (12)$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 7 (Asymptotic of β_n from the exact form). *As $n \rightarrow \infty$,*

$$\beta_n = \log n + c + O(n^{-4}) \quad (13)$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k s^{2k}}, \quad |\arg s| < \pi \quad (14)$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s| = \sqrt{\frac{1}{16} + \pi^2 n^4}$ yields

$$\operatorname{Re} \psi(s) = \log |s| + O(|s|^{-2}) = \frac{1}{2} \log(\frac{1}{16} + \pi^2 n^4) + O(n^{-4}) = 2 \log n + \log \pi + O(n^{-4}) \quad (15)$$

Therefore $\frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + i\pi n^2) = \log n + \frac{1}{2} \log \pi + O(n^{-4})$ and $\beta_n = \log n + c + O(n^{-4})$. $\square$

Definition 4 (Truncated transform). *For $T > 0$, $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \frac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt = \frac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u} du \quad (16)$$

Lemma 8 (Riemann–Siegel identity). *For $t > 0$,*

$$Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\theta(t) - t \log n) + R(t), \quad N(t) := \lfloor \sqrt{t/(2\pi)} \rfloor, \quad R(t) = O(t^{-1/4}) \quad (17)$$

Proof. [3], §7.4, Theorem 7.1. $\square$

Theorem 1 (Band-limitedness of Y). *For every $c > c_*$ and every $\mu \in \mathbb{R}$ with $|\mu| > 1$, $\lim_{T \rightarrow \infty} K_T(\mu) = 0$. Equivalently, $\operatorname{supp} S \subseteq [-1, 1]$, where $S := |\hat{Y}|^2$.*

Proof. Substituting Lemma 8 into K_T and expanding $\cos \alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$,

$$K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma \in \{\pm 1\}} \sum_{n \leq N(T)} n^{-1/2} J_{n,\sigma}(T, \mu) + \mathcal{R}(T, \mu) \quad (18)$$

with $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T, \mu) := \int_{t_n}^T \sqrt{\Theta'(t)} e^{i[\sigma\theta(t) - \sigma t \log n - \mu\Theta(t)]} dt \quad (19)$$

$$\mathcal{R}(T, \mu) := \frac{1}{2\pi} \int_0^T R(t) \sqrt{\Theta'(t)} e^{-i\mu\Theta(t)} dt \quad (20)$$

The argument proceeds in the following steps.

(i) *Remainder vanishing.* $R(t) \sqrt{\Theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$; the phase derivative $|\mu\Theta'(t)| \geq |\mu|(c - c_*)$ and grows as $\frac{|\mu|}{2} \log(t/(2\pi))$. Integration by parts gives $\mathcal{R}(\infty, \mu) - \mathcal{R}(T, \mu) = O(T^{-1/4}(\log T)^{-1/2}) \rightarrow 0$.

(ii) *Change of variable.* Set $x := \Theta'(t) = \theta'(t) + c$, $dx = \theta''(t) dt$, $X := \Theta'(T)$:

$$J_{n,\sigma}(T, \mu) = \int_{\beta_n \vee \Theta'(0)}^X \frac{\sqrt{x}}{\theta''(t(x))} e^{i\tilde{\Phi}_{n,\sigma,\mu}(x)} dx \quad (21)$$

$$\tilde{\Phi}'_{n,\sigma,\mu}(x) = \frac{(\sigma - \mu)x - \sigma \log n - \sigma c}{\theta''(t(x))} \quad (22)$$

(iii) *Exact θ'' cancellation.*

$$Q_{n,\sigma,\mu}(x) = \frac{\sqrt{x}}{(\sigma - \mu)x - \sigma \log n - \sigma c}, \quad Q'_{n,\sigma,\mu}(x) = O(x^{-3/2}) \quad (23)$$

(iv) *Finiteness of the exceptional set.* Define $\mathcal{S}(\mu) := \{n : \beta_n \leq (\log n + c)/(1 + |\mu|)\}$. By Lemma 7,

$$\beta_n - \frac{\log n + c}{1 + |\mu|} = \frac{|\mu|}{1 + |\mu|}(\log n + c) + O(n^{-4}) \rightarrow \infty \quad (24)$$

therefore $\mathcal{S}(\mu)$ is finite.

(v) *IBP for $n \notin \mathcal{S}(\mu)$.* Constant-sign phase derivative; boundary $O(X^{-1/2})$, tail $\leq 2CX^{-1/2}$.

(vi) *IBP tail for $n \in \mathcal{S}(\mu)$.* Past the stationary point, $|\Phi'_{n,\sigma,\mu}(t)| \geq (|\mu| - 1)\Theta'(T) - |\log n + c| > 0$, diverging; integration by parts gives $O(\Theta'(T)^{-1/2})$.

(vii) *Assembly.*

$$K_\infty(\mu) - K_T(\mu) = \frac{1}{2\pi} \sum_{\sigma} \left[\sum_{n \notin \mathcal{S}(\mu)} n^{-1/2} O(X^{-1/2}) + \sum_{n \in \mathcal{S}(\mu)} n^{-1/2} O(\Theta'(T)^{-1/2}) \right] + O(T^{-1/4}(\log T)^{-1/2}) \quad (25)$$

All terms vanish as $T \rightarrow \infty$; $K_\infty(\mu)$ exists.

(viii) *Identification.* The mode (n, σ) instantaneous u -frequency is

$$\sigma \cdot \frac{\theta'(t) - \log n}{\Theta'(t)} = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c} \quad (26)$$

On $t \geq 2\pi n^2$, $\theta'(t) \geq \theta'(2\pi n^2) = \beta_n - c = \frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + i\pi n^2) - \frac{1}{2} \log \pi$. Numerator ≥ 0 ; denominator exceeds numerator by $c + \log n > 0$; ratio in $[0, 1)$. The Riemann–Siegel remainder has u -frequency $O((\log t)^{-1}) \rightarrow 0$.

Therefore $\operatorname{supp} S \subseteq [-1, 1]$, and $K_\infty(\mu) = \widehat{Y}(\mu) = 0$ for $|\mu| > 1$. $\square$

6 Entire Extension and Laguerre–Pólya Membership

Corollary 1 (Paley–Wiener). *Y extends to an entire function of exponential type ≤ 1 :*

$$Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi, \quad |Y(z)| \leq \|\widehat{Y}\| e^{|\operatorname{Im} z|} \quad (27)$$

Proof. Y is a tempered distribution on $\mathbb{R}$ whose Fourier transform is supported in the compact set $[-1, 1]$. By [4], Theorem 7.3.1 (distributional Paley–Wiener), Y extends to an entire function of exponential type with type constant bounded by the support radius, which here is 1; the representation $Y(z) = \int_{-1}^1 \widehat{Y}(\xi) e^{i\xi z} d\xi$ follows from the same theorem, and $|Y(z)| \leq \|\widehat{Y}\| e^{|\operatorname{Im} z|}$ is the bound on the right. $\square$

Lemma 9 (Non-negativity of the spectral measure). *The spectral measure $S := |\widehat{Y}|^2$ is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$.*

Proof. The autocorrelation $C(\eta) := \langle Y, Y(\cdot + \eta) \rangle$ is positive-definite in the tempered sense; Bochner–Schwartz ([7], Theorem 7.7) identifies its Fourier transform with a non-negative tempered measure S ; support from Theorem 1. $\square$

Theorem 2 (Akhiezer). *A real-valued entire function F of exponential type $\leq \tau$ whose spectral measure $|\hat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F are real.*

Proof. [5], Lecture 16, Theorem 3; [2], §V.4. $\square$

Corollary 2 (Reality of zeros of Y). *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 1), with spectral measure S non-negative and supported in $[-1, 1]$ (Lemma 9). Apply Theorem 2. $\square$

7 Transfer to the Hardy Z-Function

Theorem 3 (Unitary inverse phase pullback identity on the ε -sub-strip). *Fix $\varepsilon \in (0, \frac{1}{2})$ and an ε -admissible c . Write $\Omega_\varepsilon := \{|\operatorname{Im} t| < \frac{1}{2} - \varepsilon\}$. For every $t \in \Omega_\varepsilon$,*

$$Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)} \quad (28)$$

where $\sqrt{\Theta'(\cdot)}$ is the holomorphic branch on Ω_ε that is positive on $\mathbb{R}$.

Proof. On $[0, \infty)$ the identity is Definition 2 rearranged. Both sides are holomorphic on Ω_ε : Z is holomorphic on $\Omega \supset \Omega_\varepsilon$ by Lemma 6; Θ is holomorphic on Ω_ε ; Y is entire by Corollary 1; $\sqrt{\Theta'(t)}$ admits a holomorphic branch on the open convex (hence simply connected) strip Ω_ε since Θ' has $\operatorname{Re} \Theta' \geq c - c_{**}(\varepsilon) > 0$ there (Lemma 4) and therefore is non-vanishing on Ω_ε . The identity theorem extends the equality from $[0, \infty)$ to Ω_ε . $\square$

Theorem 4 (Reality of zeros of Z in the ε -sub-strip). *Fix $\varepsilon \in (0, \frac{1}{2})$ and an ε -admissible c . Every zero of Z in Ω_ε is real.*

Proof. Let $t_0 \in \Omega_\varepsilon$ satisfy $Z(t_0) = 0$. Theorem 3 gives $Y(\Theta(t_0))\sqrt{\Theta'(t_0)} = 0$. Lemma 4 gives $\sqrt{\Theta'(t_0)} \neq 0$, therefore $Y(\Theta(t_0)) = 0$. Corollary 2 gives $\Theta(t_0) \in \mathbb{R}$. Lemma 5 on Ω_ε , therefore, gives $t_0 \in \mathbb{R}$. $\square$

Theorem 5 (Reality of zeros of Z in the full open strip). *Every zero of Z in $\Omega = \{|\operatorname{Im} t| < \frac{1}{2}\}$ is real.*

Proof. Let $t_0 \in \Omega$ satisfy $Z(t_0) = 0$. Since $|\operatorname{Im} t_0| < \frac{1}{2}$, choose $\varepsilon_0 \in (0, \frac{1}{2})$ with $|\operatorname{Im} t_0| < \frac{1}{2} - \varepsilon_0$, so $t_0 \in \Omega_{\varepsilon_0}$. Pick any ε_0 -admissible c (such c exist since $c_{**}(\varepsilon_0) < \infty$ by Lemma 3). Apply Theorem 4 to conclude $t_0 \in \mathbb{R}$. $\square$

8 The Zero Locus of ζ

Corollary 3 (Riemann hypothesis). *Every non-trivial zero ρ of the Riemann zeta function satisfies $\operatorname{Re} \rho = \frac{1}{2}$.*

Proof. Non-trivial zeros lie in $\{0 < \operatorname{Re} s < 1\}$, which under $s = \frac{1}{2} + it$ corresponds to Ω . For $\rho = \frac{1}{2} + it_0$ in Ω , $e^{-i\theta(t_0)}$ is non-vanishing and $\zeta(\frac{1}{2} + it_0) = e^{-i\theta(t_0)}Z(t_0)$, therefore $\zeta(\rho) = 0 \iff Z(t_0) = 0$. Theorem 5 gives $t_0 \in \mathbb{R}$, therefore $\operatorname{Re} \rho = \frac{1}{2}$. $\square$

A Answers to Frequently Anticipated Questions

This appendix collects short, colloquial explanations of terminology and conventions used in the main text.

A.1 Why “inverse phase pullback” and not “phase pushforward”?

The reason is semantic, not technical. The intuition is geometric, in the plain English sense of the words.

Pullback. Picture yourself standing at a point with stuff scattered across a distant landscape. You throw out strings, hook onto the stuff that is out there, and reel it back in toward where you are standing — like hauling in a kite. The object ends up concentrated at your location, retrieved from the space it was spread over. That retrieving motion is a pullback.

Pushforward. Now picture yourself sitting with a kite wound up at your feet and you want it out in the sky: you let out the line and send it forth, expanding it into the space it will occupy. That outgoing, spreading motion is a pushforward.

Apply that picture here. Z is being sent outward: it is spread across the t -line, expanded into that space. To build Y on the u -line, one sits at a point u , reaches back through the phase map via Θ^{-1} , grabs the value $Z(\Theta^{-1}(u))$, and reels it in. That retrieving is exactly why the operation is called a pullback: Z has been sent out, and one has to pull it back to gather it at u . Since the reeling is done along Θ^{-1} , the qualifier is “inverse phase.” The adjective “unitary” records that the half-Jacobian $1/\sqrt{\Theta'(\Theta^{-1}(u))}$ is included so that L^2 -mass is preserved under the operation.

Contravariance and covariance. The two directions carry standard names in differential geometry: a pullback is *contravariant* and a pushforward is *covariant*, referring to how each transforms under composition of maps. Objects that live as functions or scalar fields — things one evaluates at a point — transform contravariantly: they pull back, against the direction of the map. Objects that live as tangent vectors, velocities, or measures — things that carry a notion of flow or mass — transform covariantly: they push forward, with the direction of the map. The split is the same one that distinguishes covectors from vectors, or differential forms from densities: it records which kind of geometric object is being transported and which way the arrow of transport naturally runs. Under this classification, Z and Y are scalar-like, half-density data, and the natural transport between them is contravariant — a pullback.

A remark on the dual naming. Technically one could reverse the point of view and call Z itself the *phase pushforward* of the stationary process Y : starting from Y on the u -line, one lets out the line along Θ and expands Y outward into t -space, obtaining Z . Equivalently, the stationary process Y could be defined as the pushforward of Z along the phase — sending Z forth into u -coordinates. This naming is formally consistent, but it reads against the grain of what is being constructed. Stationarity is a property of something gathered and settled; producing it by *pushing* an object outward into a new space is the wrong English motion. One naturally assembles a stationary object by *pulling in* values from the non-stationary source. “Unitary inverse phase pullback” is therefore the name used throughout: it matches the geometric direction of the construction and the linguistic image of stationarity as something retrieved, not something broadcast.

Ditty. English grammar and category theory disagree here. In plain English a *function* is the active thing — something that acts, takes an argument, produces a value; a *measure* is what remains, passively assigning mass to sets without acting on anything. The English grammar would

therefore pair functions with pushforward and measures with pullback. Category theory goes the other way: functions transform contravariantly (they pull back), measures transform covariantly (they push forth), because a pullback of a function f along Θ is the composition $f \circ \Theta$ (the function retreats along the map), while the pushforward of a measure μ is $\mu \circ \Theta^{-1}$ (the mass is transported forward along the map). The category-theoretic convention wins in practice; the English reading is the anti-pattern. The mnemonic records the inversion:

In English a function acts, a measure stays put;
in category theory that reading's kaput:
functions pull back, measures push forth,
and the unitary inverse phase pullback points north.

A.2 Why is the spectral measure non-negative?

The autocorrelation of any real, locally square-integrable function is a positive-definite tempered distribution: the Gram matrix of finitely many translates of Y restricted to a common window has non-negative eigenvalues, a one-line consequence of Cauchy–Schwarz. Bochner–Schwartz identifies the Fourier transform of a positive-definite tempered distribution with a non-negative tempered measure, and Wiener–Khinchin identifies that Fourier transform with $S = |\widehat{Y}|^2$. Non-negativity of the spectral measure is the Fourier-side restatement of this universal property of autocorrelations; the support condition $\text{supp } S \subseteq [-1, 1]$ is delivered by Theorem 1 and combines with non-negativity to invoke Akhiezer.

A.3 What is being autocorrelated?

The autocorrelation in Lemma 9 is that of Y itself, the unitary inverse phase pullback of Z . In u -coordinates,

$$C(\eta) = \lim_{U \rightarrow \infty} \frac{1}{2U} \int_{-U}^U Y(u) Y(u + \eta) du \quad (29)$$

a positive-definite tempered distribution. Because the map $Z \mapsto Y$ is an L^2 -isometry under $u = \Theta(t)$, this is equivalent to correlating Z against itself with respect to the u -shift η , which is a nonlinear (roughly logarithmic) shift in the original t -coordinate.

A.4 What does the picture look like?

Three objects merit mental plots.

The function $Y(u)$. An envelope-modulated real oscillation of zero mean. The envelope decays like $1/\sqrt{\log t}$ expressed in u -coordinates. The zero crossings are the images under Θ of the non-trivial Riemann zeros; the phase variable straightens out their spacing.

The autocorrelation $C(\eta)$. A sharp peak at $\eta = 0$, symmetric in η because Y is real, with oscillatory decay whose frequency content sits inside $[-1, 1]$.

The spectral measure S . A non-negative tempered measure strictly supported in $[-1, 1]$, symmetric about $\xi = 0$ after the even extension. Its shape inside $[-1, 1]$ encodes the distribution of zero spacings of Z in u -coordinates.

A.5 Locally L^2 and the growing variance

Y is locally L^2 , on compact subsets $[-U, U]$ of arbitrarily large length U . The band-limit established in Theorem 1 is a statement about the Fourier support of Y as a tempered distribution on $\mathbb{R}$. All

uses of Paley–Wiener, Bochner–Schwartz, and Akhiezer in the main text are read in the tempered-distributional form, the framework for band-limited objects that are not globally L^2 .

The growing variance manifests in three compatible ways.

Envelope in u -coordinates. $Y(u)$ has an amplitude envelope that grows slowly as $|u| \rightarrow \infty$; the local mean-square $\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du$ grows without bound as $U \rightarrow \infty$.

Autocorrelation as a tempered distribution. $C(\eta)$ is the windowed-limit positive-definite distribution of §A.3. Bochner–Schwartz identifies its Fourier transform with the non-negative tempered measure S , and Theorem 1 localizes S to $[-1, 1]$.

Spectral measure as a tempered measure on $[-1, 1]$. S is a non-negative tempered measure on $\mathbb{R}$, supported in $[-1, 1]$, of infinite total mass.

Representation at finite window. Pointwise values of the spectral density at interior $\xi_0 \in (-1, 1)$ are approached through windowed periodograms

$$P_U(\xi_0) := \frac{1}{2U} \left| \int_{-U}^U Y(u) e^{-i\xi_0 u} du \right|^2 \quad (30)$$

Any figure produced at finite U is a *representation* at window scale U .

A.6 Scale of the spectral representation at finite window

Fix $\xi \in (-1, 1)$ and let $P_U(\xi)$ be the windowed periodogram. Two features of the plot change with U : the overall vertical scale, and the local fluctuation amplitude around that scale. Both are controlled by the local mean power

$$\bar{P}(U) := \frac{1}{2U} \int_{-U}^U Y(u)^2 du = \frac{1}{2U} \int_{\Theta^{-1}(-U)}^{\Theta^{-1}(U)} Z(t)^2 dt \quad (31)$$

the second equality by the L^2 -isometry of the unitary inverse phase pullback. By Parseval in the tempered sense,

$$\int_{-\infty}^{\infty} P_U(\xi) d\xi = 2\pi \bar{P}(U) \quad (32)$$

Using $\Theta(t) \sim \frac{t}{2} \log(t/(2\pi e)) + ct$ and the Hardy–Littlewood moment $\int_0^T Z(t)^2 dt \sim T \log T$ ([8], §7.3),

$$\bar{P}(U) \sim \log(\Theta^{-1}(U)) \sim \log U + \log \log U + O(1) \quad (U \rightarrow \infty) \quad (33)$$

$P_U(\xi)$ fluctuates around $\bar{P}(U)$ on a scale comparable to $\bar{P}(U)$ itself, with wider fluctuations near the edges $|\xi| \rightarrow 1^-$.

A.7 The normalized spectral measure

At every finite U the periodogram P_U has finite total mass $2\pi \bar{P}(U)$ by Parseval. Dividing by this total mass produces the normalized spectral probability measure at scale U ,

$$\tilde{P}_U(\xi) := \frac{P_U(\xi)}{2\pi \bar{P}(U)}, \quad \int_{-\infty}^{\infty} \tilde{P}_U(\xi) d\xi = 1 \quad (34)$$

a probability density on $\mathbb{R}$ at every U . By Theorem 1 the probability mass of $\tilde{P}_U$ outside $[-1, 1]$ vanishes as $U \rightarrow \infty$, so the family is tight on $[-1, 1]$. By Helly–Prokhorov, $\tilde{P}_U$ converges weakly as

$U \rightarrow \infty$ to a probability measure ν_∞ on $[-1, 1]$:

$$\int f d\tilde{P}_U \rightarrow \int f d\nu_\infty \quad \text{for every } f \in C_b(\mathbb{R}) \quad (35)$$

ν_∞ is the normalized spectral density of Y : a probability measure of total mass 1 supported in $[-1, 1]$.

S and ν_∞ are distinct, both well-defined:

- S : non-negative tempered spectral measure on $[-1, 1]$, infinite total mass, delivered by Theorem 1 and Bochner–Schwartz ([7], Theorem 7.7).
- ν_∞ : probability measure on $[-1, 1]$, the weak-* limit of the normalized windowed periodograms $\tilde{P}_U$.

A.8 Cramér representation of the sample path

As a generalized second-order stationary process on $\mathbb{R}_u$ with non-negative tempered spectral measure S supported in $[-1, 1]$, Y admits a Cramér spectral representation of its sample path,

$$Y(u) = \int_{-1}^1 e^{i\xi u} d\Phi(\xi) \quad (36)$$

where $d\Phi$ is a complex-valued orthogonal Borel measure on $[-1, 1]$ with covariance structure

$$\mathbb{E}[d\Phi(\xi) \overline{d\Phi(\xi')}] = \delta(\xi - \xi') dS(\xi) \quad (37)$$

The support condition $\text{supp } S \subseteq [-1, 1]$ of Theorem 1 localizes $d\Phi$ to the interval $[-1, 1]$. Reality of Y on $\mathbb{R}$ is equivalent to the Hermitian symmetry $d\Phi(-\xi) = \overline{d\Phi(\xi)}$, under which S is an even measure on $[-1, 1]$. Test functions f are paired with $d\Phi$ under the $L^2(S)$ isometry

$$\mathbb{E} \left| \int f(\xi) d\Phi(\xi) \right|^2 = \int |f(\xi)|^2 dS(\xi) \quad (38)$$

which requires $f \in L^2(S)$ and permits S to have infinite total mass. The Cramér representation exists at the level of S : it is the reconstruction theorem for the process. ν_∞ is the scale-invariant spectral shape of the same process.

Inverse representation of $d\Phi$. The orthogonal Borel measure $d\Phi$ is recovered from the sample path Y by the truncated Fourier inversion

$$\Phi(B) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \widehat{\mathcal{K}}_B(-u) du \quad (39)$$

for every Borel set $B \subseteq [-1, 1]$, the limit taken in $L^2(\Omega_P, \mathbb{P})$; here $\widehat{\mathcal{K}}_B(u) = \int_B e^{i\xi u} d\xi$. Equivalently, for every $f \in L^2(S)$,

$$\int_{-1}^1 f(\xi) d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{2\pi} \int_{-U}^U Y(u) \check{f}(u) du, \quad \check{f}(u) = \int_{-1}^1 f(\xi) e^{-i\xi u} d\xi \quad (40)$$

again in $L^2(\Omega_P, \mathbb{P})$. Existence of the limit is the $L^2(S)$ isometry (38) together with the truncated Fourier inversion on Y .

Inserting $d\Phi$ into the forward representation. Fix $c > c_*$ and $t \in \mathbb{R}$. Apply (40) with test function $f(\xi) = e^{i\xi\Theta(t)}$; this f is continuous on $[-1, 1]$ hence in $L^2(S|_{[-1,1]})$. The inner ξ -integral in the inverse formula is Lebesgue,

$$\check{f}(u) = \int_{-1}^1 e^{i\xi\Theta(t)} e^{-i\xi u} d\xi = \int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi = \frac{2 \sin(\Theta(t) - u)}{\Theta(t) - u} \quad (41)$$

so that (40) reads

$$\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) = \lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du \quad (42)$$

in $L^2(\Omega_P, \mathbb{P})$. The passage from the nested (ξ, u) -form of (40) to the single u -integral (42) exchanges the order of ξ - and u -integration. At every finite U this is Fubini on the product measure $d\xi \otimes du$ on $[-1, 1] \times [-U, U]$: the integrand satisfies

$$|Y(u) e^{i\xi(\Theta(t)-u)}| = |Y(u)| \quad (43)$$

and

$$\int_{-U}^U \int_{-1}^1 |Y(u)| d\xi du = 2 \int_{-U}^U |Y(u)| du < \infty \quad (44)$$

since Y is locally L^2 on $\mathbb{R}$ hence locally L^1 on the compact $[-U, U]$; $[-1, 1]$ has finite Lebesgue measure. Fubini therefore applies: the double integral factors in either order, with the same value,

$$\int_{-U}^U Y(u) \left(\int_{-1}^1 e^{i\xi(\Theta(t)-u)} d\xi \right) du = \int_{-1}^1 \left(\int_{-U}^U Y(u) e^{i\xi(\Theta(t)-u)} du \right) d\xi \quad (45)$$

and the $L^2(\Omega_P, \mathbb{P})$ -limit $U \rightarrow \infty$ on both sides exists, with the limit on the right equal to $\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$ by truncated Fourier inversion for Y . The bound (43) is independent of c , and the only appearance of c in (45) is through the real translate $\Theta(t) = \theta(t) + ct$ in the phase; Fubini's hypothesis is therefore satisfied for every $c > c_*$.

Time-warped representation of Z . The limit on the right of (42) is the sinc-kernel sampling of Y at the point $\Theta(t) \in \mathbb{R}$. Since Y is band-limited to $[-1, 1]$ by Theorem 1 and $\Theta(t) \in \mathbb{R}$, Paley–Wiener sampling gives

$$\lim_{U \rightarrow \infty} \frac{1}{\pi} \int_{-U}^U Y(u) \frac{\sin(\Theta(t) - u)}{\Theta(t) - u} du = Y(\Theta(t)) \quad (46)$$

in $L^2(\Omega_P, \mathbb{P})$. Combining (42), (46), and Theorem 3 ($Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$),

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)) = \sqrt{\Theta'(t)} \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \quad (47)$$

for every $c > c_*$ and every $t \in \mathbb{R}$. This is the unitary time-change of the Cramér representation: the same orthogonal Borel measure $d\Phi$ that represents the stationary sample path Y represents Z after the reparametrisation $u \mapsto \Theta(t)$ and the half-Jacobian amplitude $\sqrt{\Theta'(t)}$. The representation (47) depends on c only through $\Theta = \theta + c \cdot \text{id}$; the orthogonal Borel measure $d\Phi$ itself is a spectral object of Y , independent of c .

Covariance of Z . From (47), the covariance of Z at $t, s \in \mathbb{R}$ is

$$R_Z(t, s) := \mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \int_{-1}^1 e^{i\xi(\Theta(t) - \Theta(s))} dS(\xi) \quad (48)$$

Derivation of (48) from (47). Fix $t, s \in \mathbb{R}$. Expanding the product and pulling the amplitudes outside the expectation,

$$\mathbb{E}[Z(t) \overline{Z(s)}] = \sqrt{\Theta'(t) \Theta'(s)} \mathbb{E} \left[\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] \quad (49)$$

The two Wiener integrals are each elements of $L^2(\Omega_P, \mathbb{P})$ with $L^2(S)$ -norms equal to $\|e^{i\cdot\Theta(t)}\|_{L^2(S)} = S([-1, 1])^{1/2}$ and $\|e^{i\cdot\Theta(s)}\|_{L^2(S)} = S([-1, 1])^{1/2}$, which are finite whenever the integrals are taken against a compactly supported test function — here the restriction of $e^{i\cdot\Theta(t)}$ and $e^{i\cdot\Theta(s)}$ to the compact set $[-1, 1]$. The Wiener–Itô product rule (the bilinear form of (38)) gives

$$\mathbb{E} \left[\int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi) \overline{\int_{-1}^1 e^{i\xi'\Theta(s)} d\Phi(\xi')} \right] = \int_{-1}^1 e^{i\xi\Theta(t)} \overline{e^{i\xi\Theta(s)}} dS(\xi) \quad (50)$$

Combining (49) and (50) gives (48).

The exchange of the two ξ -integrals with the expectation in (50) is an instance of the bilinear Itô isometry and not an application of Fubini directly. The Fubini justification of that bilinear isometry is standard: for simple functions $f = \sum_k a_k \mathbb{1}_{B_k}$ with $B_k \subseteq [-1, 1]$ pairwise disjoint and Borel, and $g = \sum_\ell b_\ell \mathbb{1}_{C_\ell}$ similarly, orthogonality of the increments $\{\Phi(B_k)\}$ gives

$$\mathbb{E} \left[\int f d\Phi \overline{\int g d\Phi} \right] = \sum_k \sum_\ell a_k \overline{b_\ell} \mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = \sum_k a_k \overline{b_k} S(B_k \cap C_k) = \int f \overline{g} dS \quad (51)$$

where in the middle equality we used $\mathbb{E}[\Phi(B_k) \overline{\Phi(C_\ell)}] = S(B_k \cap C_\ell)$ (the integrated form of (37)) and the fact that for simple functions the double sum is finite so Fubini applies pointwise. Bounded-continuous functions on $[-1, 1]$ are $L^2(S)$ -limits of simple functions (standard, since $S([-1, 1]) < \infty$ if one replaces S by S -on- $[-1, 1]$, which is finite because Theorem 1 supports S in $[-1, 1]$ but does not bound its total mass; for the identity (50) the tests $e^{i\cdot\Theta(t)}$ and $e^{i\cdot\Theta(s)}$ are bounded on the compact set $[-1, 1]$, so they are in $L^2(S|_{[-1, 1]})$ even when S itself is infinite on $\mathbb{R}$), and the bilinear form in (50) extends continuously in both arguments by the $L^2(S)$ isometry (38), yielding the claim. No assumption on c is used beyond $c > c_*$, which is the standing hypothesis on Θ .

The exchange in (47) itself — the equality $\sqrt{\Theta'(t)} Y(\Theta(t)) = \sqrt{\Theta'(t)} \int_{-1}^1 e^{i\xi\Theta(t)} d\Phi(\xi)$ — is the pointwise substitution $u \mapsto \Theta(t)$ in (36), valid because $\Theta(t) \in \mathbb{R}$ for $t \in \mathbb{R}$ and the sample path $Y(\Theta(t))$ is defined by the Cramér representation evaluated at the point $\Theta(t)$. No integral exchange is required: the inner Wiener integral is the same element of $L^2(\Omega_P, \mathbb{P})$, with the phase $e^{i\xi u}$ evaluated at $u = \Theta(t)$. $\square$

The identity (48) is the time-warped autocovariance of the stationary underlying process. Non-stationarity of Z is expressed entirely through the pair $(\Theta'(t)\Theta'(s))^{1/2}$ and the argument $\Theta(t) - \Theta(s)$: the frequency content is the same S supported in $[-1, 1]$; the time dependence is absorbed into the biholomorphic reparametrisation Θ and the half-Jacobian amplitude.

A.9 Variance structure function

At coincident times $s = t$ the covariance (48) reduces to the variance of $Z(t)$, and the ξ -integral in (48) loses its oscillatory factor. Write

$$\sigma_Y^2 := S([-1, 1]) = \int_{-1}^1 dS(\xi) \quad (52)$$

for the total mass of S restricted to $[-1, 1]$; this quantity is finite when Y is second-order on $\mathbb{R}$, which is the hypothesis of the Cramér representation (36). Setting $s = t$ in (48) then gives $\Theta(t) - \Theta(s) = 0$, the exponential equals 1, and

$$\text{Var } Z(t) = R_Z(t, t) = \Theta'(t) \int_{-1}^1 dS(\xi) = \sigma_Y^2 \Theta'(t) \quad (53)$$

for every $t \in \mathbb{R}$. The variance of Z is the total spectral mass of Y amplified by the instantaneous phase rate $\Theta'(t)$; it is the diagonal restriction of the covariance, with no dependence on the off-diagonal phase.

Pinning of σ_Y^2 by the Hardy–Littlewood second moment. The value $\sigma_Y^2 = 2$ is not a free calibration; it is pinned as a theorem by the Hardy–Littlewood second moment of ζ . Integrating (53) against dt on $[0, T]$,

$$\int_0^T \text{Var } Z(t) dt = \sigma_Y^2 \int_0^T \Theta'(t) dt = \sigma_Y^2 [\Theta(T) - \Theta(0)]. \quad (54)$$

On the left, using $|Z(t)| = |\zeta(\frac{1}{2} + it)|$ on $\mathbb{R}$ (from $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$ with $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$),

$$\int_0^T \text{Var } Z(t) dt = \int_0^T \mathbb{E} |\zeta(\frac{1}{2} + it)|^2 dt = \int_0^T |\zeta(\frac{1}{2} + it)|^2 dt, \quad (55)$$

the outer expectation being trivial since $|\zeta(\frac{1}{2} + it)|^2$ is not a chance quantity. On the right, $\Theta(T) = \theta(T) + cT = \frac{T}{2} \log(T/(2\pi)) - \frac{T}{2} + cT + O(1)$. Hardy–Littlewood ([8], §7.3) gives $\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log T - (1 + \log 2\pi - 2\gamma)T + O(T^{1/2} \log T)$. Dividing both (54) and the Hardy–Littlewood identity by $T \log T$ and taking $T \rightarrow \infty$ equates the two leading coefficients: the right side of (54) contributes $\sigma_Y^2 \cdot \frac{1}{2}$ and Hardy–Littlewood contributes 1, forcing

$$\sigma_Y^2 = 2. \quad (56)$$

No degree of freedom remains; $\sigma_Y^2 = 2$ is a theorem derived from Hardy–Littlewood, not a convention. With $\sigma_Y^2 = 2$, (53) becomes the exact identity

$$\text{Var } Z(t) = 2 \Theta'(t), \quad (57)$$

with $\Theta'(t) = \theta'(t) + c = \frac{1}{2} \text{Re } \psi(\frac{1}{4} + \frac{it}{2}) - \frac{1}{2} \log \pi + c$ by (3). No asymptotic expansion of θ' enters: (57) holds for every $t \in \mathbb{R}$ as an identity between $\text{Var } Z$ and the digamma function.

A.10 Sample-path autocorrelation under the unitary time change

The stationary process Y carries a sample-path autocorrelation, and the unitary time change expresses it directly on Z . For any window $[A, B] \subset \mathbb{R}_u$ in the u -variable and any lag $\eta \in \mathbb{R}$,

$$C_Y(\eta; A, B) := \int_A^{B-\eta} Y(u) \overline{Y(u+\eta)} du \quad (58)$$

is the unnormalized sample autocorrelation of Y on $[A, B]$ at lag η . Apply the substitution $u = \Theta(\tau)$, $du = \Theta'(\tau) d\tau$, using $Y(u) = Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ from (11). The pair $(u, u + \eta)$ becomes $(\Theta(\tau), \Theta(\tau) + \eta)$, and the second argument is the image of $\Theta^{-1}(\Theta(\tau) + \eta)$ under Θ . Writing $\tau'(\tau, \eta) := \Theta^{-1}(\Theta(\tau) + \eta)$ for the lagged time in t -coordinates,

$$C_Y(\eta; A, B) = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B-\eta)} Z(\tau) \overline{Z(\tau'(\tau, \eta))} \sqrt{\frac{\Theta'(\tau)}{\Theta'(\tau'(\tau, \eta))}} d\tau \quad (59)$$

where the integrand combines the two half-Jacobian factors $1/\sqrt{\Theta'(\tau)} \cdot 1/\sqrt{\Theta'(\tau'(\tau, \eta))}$ from the two evaluations of Y with the single Jacobian $\Theta'(\tau)$ from $du = \Theta'(\tau) d\tau$, giving the asymmetric ratio $\sqrt{\Theta'(\tau)/\Theta'(\tau'(\tau, \eta))}$. Equation (59) is the t -variable expression of the stationary sample-path autocorrelation of Y at lag η : translation-invariant on the u -axis becomes non-translation-invariant on the t -axis precisely through the asymmetric ratio and the Θ^{-1} -images of the window endpoints. At zero lag, $\tau'(\tau, 0) = \tau$, the ratio is 1, and

$$C_Y(0; A, B) = \int_A^B |Y(u)|^2 du = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} |Z(\tau)|^2 d\tau = \int_{\Theta^{-1}(A)}^{\Theta^{-1}(B)} \left| \zeta\left(\frac{1}{2} + i\tau\right) \right|^2 d\tau \quad (60)$$

using $|Z(\tau)| = |\zeta(\frac{1}{2} + i\tau)|$ on the real line. The right-hand side is the integrated second moment of ζ on the critical line over the window $[\Theta^{-1}(A), \Theta^{-1}(B)]$; as the window expands, it returns the Hardy–Littlewood asymptotic $T \log T$ and couples directly to (53) through $\sigma_Y^2 = 2$. Identity (59) is independent of c only up to the c -dependence of Θ : changing c rescales the window in t -coordinates through Θ^{-1} but does not change the integrated identity (60).

References

- [1] M. Abramowitz and I. A. Stegun (eds.), *Handbook of Mathematical Functions*, Dover, 1972.
- [2] N. I. Akhiezer, *Theory of Approximation*, Ungar, 1956.
- [3] H. M. Edwards, *Riemann's Zeta Function*, Academic Press, 1974.
- [4] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, 2nd ed., Springer, 1990.
- [5] B. Ya. Levin, *Lectures on Entire Functions*, Translations of Mathematical Monographs, vol. 150, American Mathematical Society, 1996.
- [6] W. Rudin, *Real and Complex Analysis*, 3rd ed., McGraw–Hill, 1987.
- [7] W. Rudin, *Functional Analysis*, 2nd ed., McGraw–Hill, 1991.
- [8] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.